

6. MASS AND BALANCE

6.1 EQUIPMENT LIST ENTRIES

System/Equipment Designation	Part No.	Mass (kg)	Arm (mm)	Massmoment (kgmm)
Aerial Flare System	105-84201	7.6	3500	26600
– Installation Provisions	105-84203	0.8	2400	1920
– Detachable Items	105-84205	6.8	3600	24480
NOTE Detachable items comprise the mount assembly and the control panel.				

7. SYSTEM DESCRIPTION

The aerial flare system provides a means for mounting and firing two aerial flare cartridges (not supplied with the system). The flares are designed to provide ground area-coverage illumination during night search and rescue operations and emergency landings.

The system comprises a dual-flare mount assembly attached to the landing-gear RH rear cross tube, a center console mounted control panel and associated wiring.

The control panel is fitted with two toggle switches; a two-position PWR switch provides 28V DC power to the system when placed in the ON position, and a three-position FIRE switch (spring-loaded in the center OFF position) supplies electrical power to ignite the selected flare cartridge. A FLARE circuit breaker on the switch panel provides system circuit protection.

The flare cartridges, each comprised of a flare candle with parachute, propellant charge, tracer unit and electrical igniter, are packed in sealed launch tubes. These tubes are inserted in sockets provided on the mount assembly and secured through use of knurled rings. The mounting sockets are covered with protective discs when not fitted with flare cartridges.

Upon firing, the flare candle with parachute is propelled out of its launch tube container to a horizontal distance of approx 15 m. The tracer unit marks the flight path of the flare candle until its ignition 2-3 seconds after firing. Flare candle burn time is 80 seconds with an approx 600 fpm rate-of-descent. At a height of 1200 ft AGL a flare will illuminate an area of 950 m in diameter.